

Mitigating Mobile Money Services frauds in Rwanda

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Abstract—For the past twelve years, Rwanda has registered a continuous increase in mobile money services subscribers. The latter along with the increased volume of money transacted has led to a proportional increase in the number of fraudulent activities targeting these users which have left many victims. This work analyzes several mobile money fraud detection techniques and presents a high-precision, high-accuracy machine learning model which can be used as a sustainable security layer that maintains the mobile money user experience while mitigating fraudulent transactions within the Mobile Money Systems.

Keywords— *Mobile Money, Fraud detection, Mobile Money Transfer Fraud, Mobile Money Attacks, Machine Learning*

I. INTRODUCTION

A considerable number of people use mobile money in Rwanda and the number has been increasing exponentially. According to an MTN report, Rwanda's mobile payment transactions grew by 206% and the active mobile money user base had grown from 2.8 million to 6.4 million in 2021 [1][2].

The mobile money transfer system is extremely security-sensitive, and because it is rapidly expanding in Rwanda, it attracts attackers and fraudsters. Mobile Money Transfers (MMT) can be used for money laundering and unlawful funding if the security is not adequately developed and maintained.

A. Definition of Terms

Fraud: Wrongful or criminal deception intended to result in financial or personal gain.

Fraud detection: Set of activities and techniques undertaken to prevent money or property from being obtained through false pretenses.

Mobile Money: An electronic wallet service that allows users to store, send and receive money using their mobile phones.

Mobile Money Attacks: Deliberate and malicious attempts to breach the security of the mobile money system to get information.

Mobile Money Transfer Fraud: Intentional and deliberate action undertaken by players in the mobile financial services ecosystem aimed at deriving gain (in cash or e-money), and/or denying other players revenue while sending and receiving money using a mobile phone [3].

Machine Learning: Development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyze and draw inferences from patterns in data [16].

B. Research Question

In this research, we aim to answer this question: *Can we identify the most efficient mobile money fraud detection technique which we can use to build an additional security layer while maintaining the user's experience?* The paper examines various existing mobile fraud detection techniques and attempts to relate them to Rwanda's current MMT system. We will look into the (i) *Existing mobile money fraud detection techniques*. (ii) *Existing infrastructure used by mobile money systems*. Using machine learning classification, we intend to determine the best technique for fraud detection based on accuracy and precision metrics. In addition, we suggest an extra security layer for Mobile Money Systems.

The audience of this study includes but is not limited to the following: Mobile Money Operators (MMO) in Rwanda, as well as policymakers who regulate security in digital transactions, particularly mobile money transactions.

C. Scope

Our research focuses primarily on MMOs in Rwanda by using MTN as a case study, and we are limited in suggesting existing best-fit solutions to Rwandan mobile money providers for the mitigation of Mobile Money Transfer fraud and attacks. However, we will not be implementing a layer at any mobile operator. It's up to the operators to integrate our solution into their services; Only then can the users benefit. Due to limited information at the MMO, our study focuses solely on detecting frauds that demonstrate a pattern of irregularities. Therefore, fraudulent transactions that seem benign will not be detectable by our proposed technique.

D. Assumptions

Given that Mobile Money Operators in Rwanda conduct day-to-day money transactions, we assume that they are already using security measures to mitigate MMT fraud and attacks.

II. IMPORTANCE AND PRIOR WORK

Mobile Money has led to an increase in financial inclusion in low- and middle-income countries, particularly in Sub-Saharan Africa with a total number of registered accounts close to 400 million at the end of 2018 [4]. Mobile money enables mobile phone owners to deposit, transfer, and withdraw funds without owning a bank account [5]. The essence of mobile money is to make different transfers from one account to another. Mobile Money Transfer refers to a financial service offered by Mobile Network Operators that allows funds transfer using electronic money through mobile channels. They can handle a large number of transactions along with the ability to move funds locally and regionally. As mobile payments begin to scale in many markets and new products are introduced, there is a growing need to address the associated fraud conclusively since this has become a major concern impacting both the mobile money service providers as well as their customers. As of 2020, 548 million registered accounts were recorded in Sub-Saharan Africa with a transaction value of 490 billion US dollars [6]. As a result of an increase in mobile money transactions, there is a proportional increase in fraudulent activities with countries like South Africa recording 220,727 incidents followed by Uganda with 68,234 incidents, and Kenya with 27,172 incidents among others. These incidents led to estimated losses of 573 million dollars, 35 million dollars, and 175 million dollars respectively [7].

In Rwanda, MTN was the first telecom company to introduce mobile money in 2010, followed by Tigo in 2011 and Airtel in 2013 [8]. Since then, MTN has registered 6.2 million subscribers and 5 million subscribers for Airtel (after merging with Tigo) as of 2021 [2][9]. The increased number of subscribers also brought with it high fraudulent activity rates that led to an estimated loss of RWF 12 million with 80 recorded incidents in just 2 months during the pandemic according to the Rwanda Investigation Bureau (RIB) [10].

Over the years different research has been carried out to find the best way to mitigate fraud in mobile money services. F. Kappelin et al [11] presented a mathematical approach to mobile money fraud detection that combined an accounting prevalent Benford's law together with quantile statistics while working with data from a fully deployed mobile money system. Other research has tested/suggested several fraud detection techniques like cryptographic functions (e.g., asymmetric encryption function, symmetric encryption function, and hash function), personal identification (e.g., number-based, and biometric-based countermeasures), and machine learning algorithms [12]. For the latter, E. Oluwatosin investigates the influence of upsampling, downsampling, and Synthetic Minority Oversampling (SMOTE) on five Machine Learning classification techniques namely decision trees, Naive Bayes, k-nearest neighbors, gradient boosted trees, and logistic regression [13].

While the reviewed literature provides tested/suggested techniques, our main goal is to determine which one would be more suited to use for building additional security to MTN's existing mobile money systems which can later be adopted by other Mobile Money Operators in Rwanda without affecting the users' experience as well as subsequently reducing the kind of frauds faced by these users.

III. METHODOLOGY

As far as we know, current MMOs use PIN authentication to accept transactions on user accounts. In addition, some MMOs increase the security by only allowing agent-initiated withdrawal after the user approves the session. As can be

observed in Fig 1, we propose an additional security layer for fraud detection on the existing Mobile Money security architecture. To find the most efficient and reliable framework, we explore machine learning-based methods.

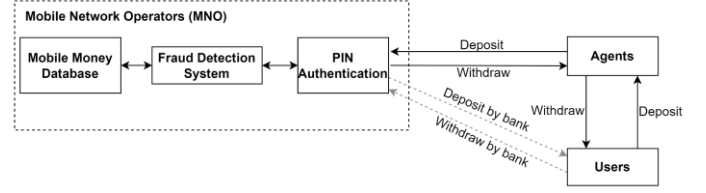


Fig 1: Fraud detection security layer

A. Data

Due to a scarcity of authentic mobile money transaction datasets on which to conduct research in the domain of fraud detection, we synthesized data using the PaySim simulator. Furthermore, we used genuine transactions collected from an African country while setting up our simulation [14]. As a result of the simulation, a new dataset with statistical attributes similar to those of the original was generated.

For each transaction record, the dataset defined the following features: type of transaction, amount of transaction, old origin balance, new origin balance, old destination balance, new destination balance, and whether it is or isn't fraudulent.

The simulator offered flexibility in defining our own target dataset distribution. To come up with parameters to use in the simulator, we first consulted MTN's terms and conditions [15], and recent quarterly reports [2]. Due to the simulation's high computational requirements for a full dataset comparable to that of MTN's 6.7m subscribers, we settled with a subsection comprising seventy thousand clients and five thousand merchants. This coupled with an assumption of a 0.01% fraud rate became the basis of our simulation parameters described in Table 1.

Simulator Parameter	Value
Timeframe	1-month
Fraud prevalence	0.01%
Number of fraudsters	2300
Number of clients	70000
Number of banks	16
Number of merchants	5000
Transfer limits	4M Rwf

Table 1: Parameters used in the dataset simulator.

To pre-process our synthesized data, we dropped nonsignificant columns that were not helping in determining the dependent variable: **isfraud**, we removed outliers and encoded categorical data to ensure our dataset is compliant with the techniques explored in the study.

B. Model

This study used machine learning classification methods to determine the best technique that should be used by mobile money providers to improve their security.

We explored 5 main classical machine learning techniques: Logistic Regression, K-Nearest Neighbor, Naïve Bayes, Random Forest, and Gradient boosted trees. Bayes, Random Forest, and Gradient boosted trees. For deep learning, we used vanilla Recurrent Neural Networks (RNN) and Long Short-Term Memory Networks (LSTM) since we are dealing with sequences of transactions.

To compare our models, we validate each technique by computing the accuracy and precision on our test dataset. Using T & F as True & False, and N & P as Negative & Positive. We defined accuracy and precision using the occurrences of TP (True Positive), TN (True Negative), FP (False Positive), and FN (False Negative) in our test set.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

In this study, our goal is to obtain a model that has an accuracy higher than 95% and a precision higher than 97%.

IV. RESULTS AND DISCUSSION

Model Name	Classification Accuracy	Precision
Logistic Regression	99.889618	0.586
K Nearest Neighbours	99.990643	0.967
Random Forest	99.999561	0.998
Gradient Boosting	99.999122	0.997
Gaussian Naive Bayes	98.965189	0.103
LSTM	99.764876	0.452
RNN	99.717962	0.322

Table 2: Different Models and their performance on the test set.

A. Results

During data pre-processing, we discovered that of the five types of transactions: cash in, cash out, transfer, debit, and payment, only two of them, cash out and transfer, are subjected to fraud. As a result, we've dropped the other types of transactions in favor of focusing on these two. Furthermore, we discovered that there is a mathematical mismatch between the amount of money deducted from the sender and the amount of money credited to the receiver's account by looking at the transaction amount and the initial balance in the sender and receiver accounts. We recognized this error as a valuable parameter for predicting the existence of fraud in the system, so we computed it and included it in our data.

After training our proposed models on the mobile money transactions, we were able to achieve decent validation accuracy and precision on the test set. In fact, the Random Forest classifier achieved a 99.9995% accuracy and a precision of 0.998, which was the highest among all the 7 models reported in Table 2.

B. Discussion

Although our primary goal was to develop a pipeline that can achieve a 95% accuracy, the assumption was that the operator would handle the false positives related to that security layer. With 95% accuracy, the rate of false positives would be as high as five in every 100 transactions. The approach presented achieved a high accuracy comparable to 1 false positive in two hundred thousand transactions. At this rate, the Mobile money operator would have improved the protection of the clients while also not overwhelming the customer support team with unnecessary complaints.

As seen, our models perform almost similarly on the accuracy metric, but major differences are noticeable when we take precision into account. From our results, the Random forest has the highest accuracy and precision, this can be attributed to its ensemble nature which averages predictions from a collection of predictor trees, thus reducing overfitting. This can also be observed in Gradient boosting, another ensemble technique that performed better than the rest.

V. CONCLUSION AND FUTURE WORK

In this paper, we present a high-precision, high-accuracy model that could be used as a security layer to mitigate fraudulent transactions on MMO systems. The focal point of our approaches was using the collective dataset of previous fraud transactions to learn the patterns that fraudulent transactions share. By the same token, we have demonstrated that good fraud detection accuracy is not enough, the rate of false-positive and false-negative may result in severe complaints from users whose benign transactions were blocked. Consequently, we optimized our model to increase the precision to 0.998. Lastly, we described how MMO can leverage the proposed models without any user experience change.

Although we were able to develop a high precision model, we heavily relied on patterns in the entirety of the mobile money transactions dataset. Consequently, our neural networks performed similarly or worse than the other machine learning techniques explored. One thing that may be considered to improve the accuracy and precision in future work is leveraging patterns in the information in individual subscriber transactions. In addition, LSTM models may be improved by restructuring the dataset to group by subscribers while also keeping temporal information intact (the order in which transactions were carried out).

Moreover, this study is limited to the PaySim synthetic dataset structure, future works may request actual MMO transaction data to develop a more realistic fraud detection model. Also, the security layer and the techniques explored may only be efficient in certain types of frauds. Frauds that may not result in account balance inconsistencies, and benign-looking fraudulent transactions may go through our proposed security undetected.

Finally, a simulated study may also be set up that mocks an MMO system and integrates our best machine learning model: the Random Forest classifier as an additional security layer for fraud detection. Testing would allow researchers to see how normal user transactions would be blocked or accepted by the systems.

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